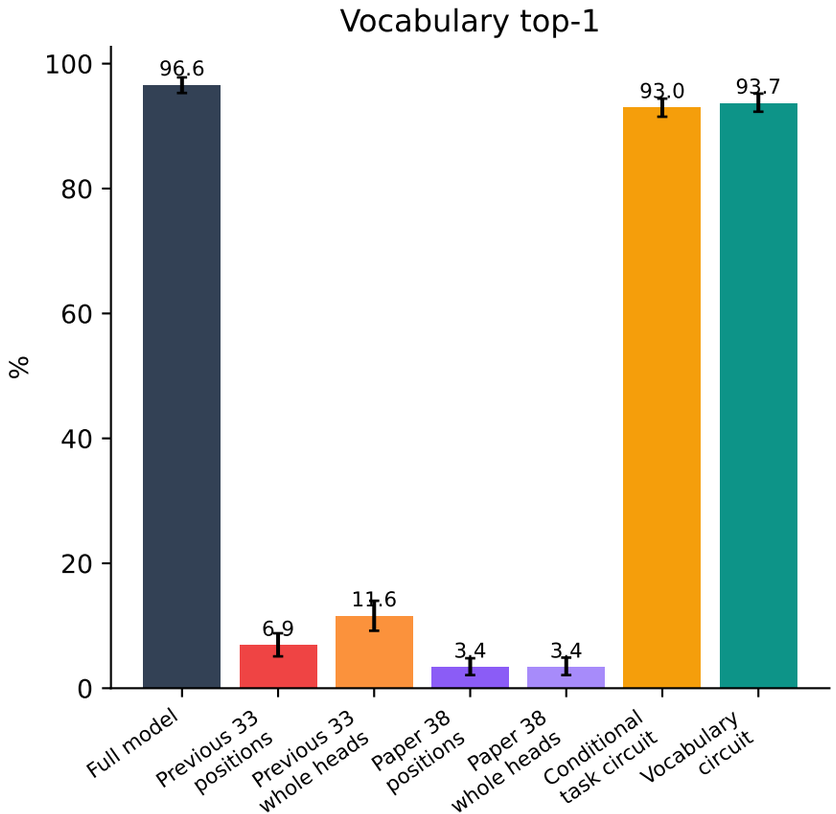
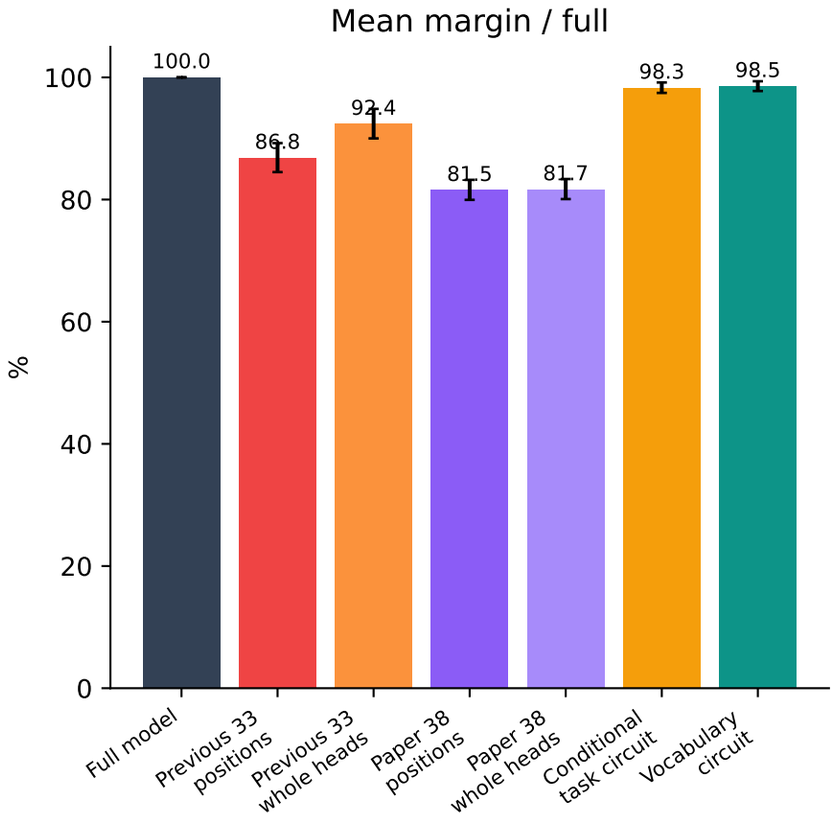
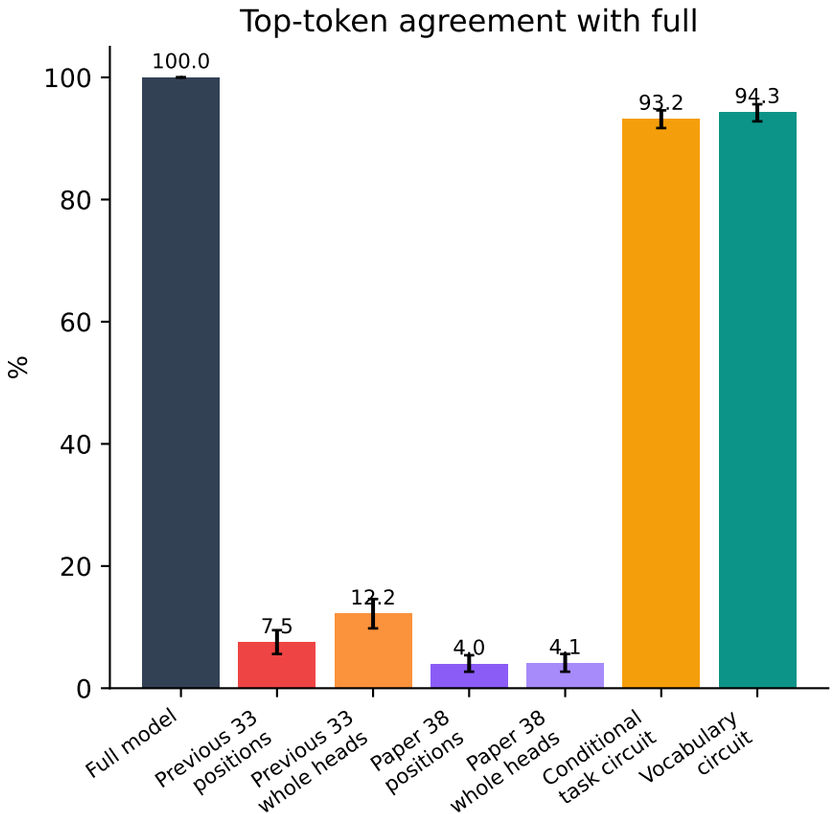
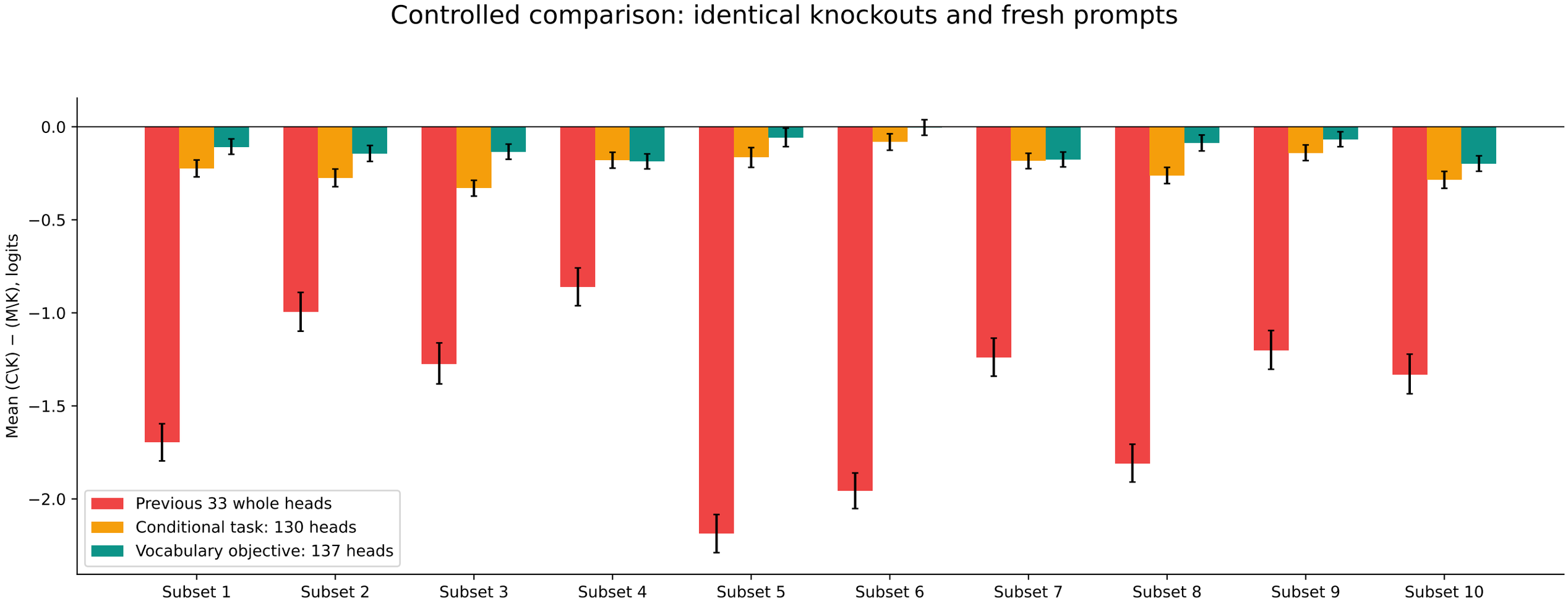
Second attempt: 137 whole heads, fresh 1,000-prompt test





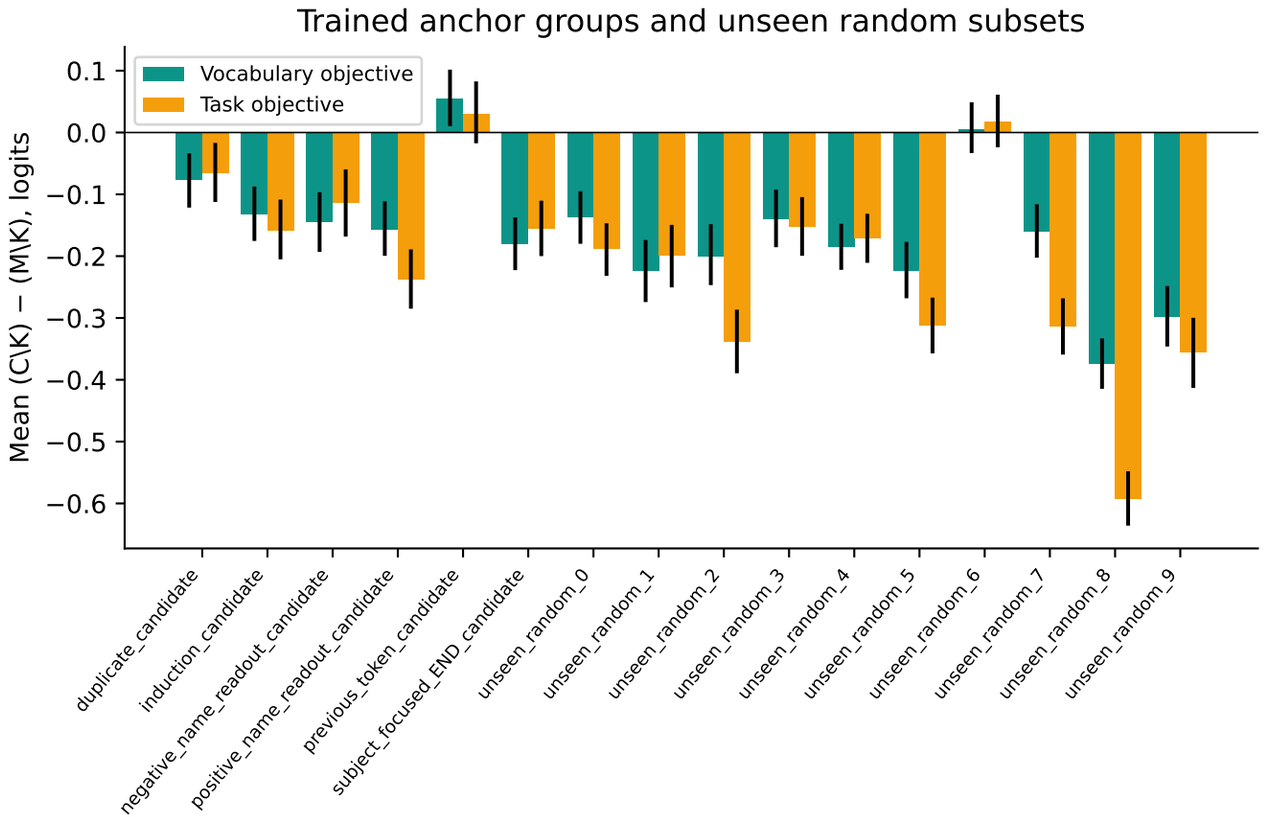


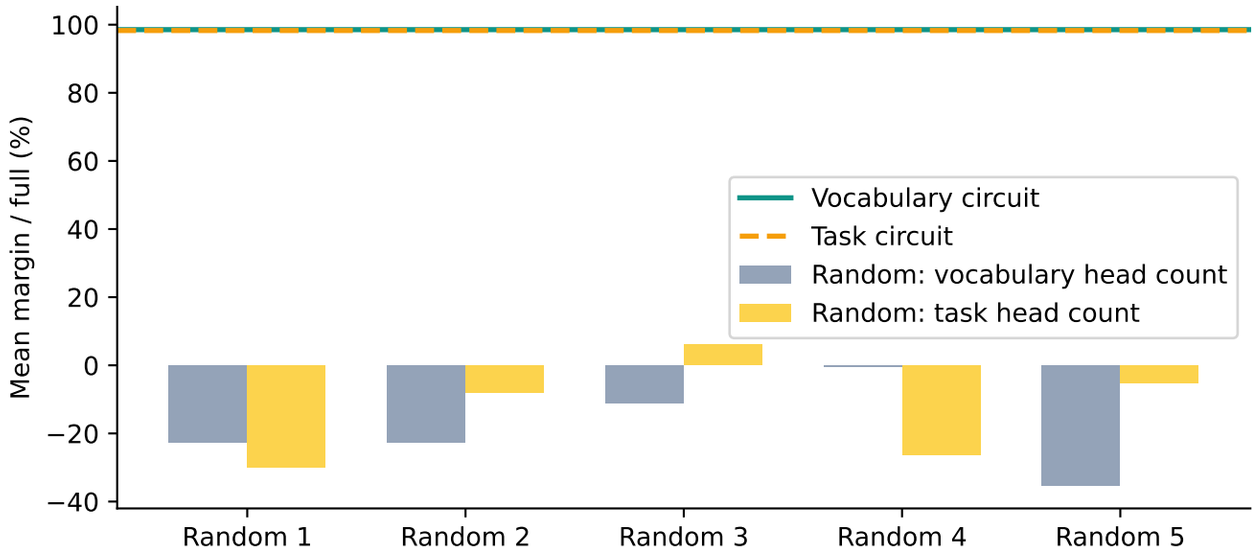
Whole heads retain all token positions. Positional masks retain selected output positions only. Same prompts and ABC reference bank. 95% matched-context bootstrap intervals; zero observed error can yield degenerate intervals.



Each subset removes 10 whole heads shared by all three masks. Same reference means and full-model knockout comparator. 95% matched-context bootstrap intervals. Covers shared subsets only; it is not an exhaustive completeness test.

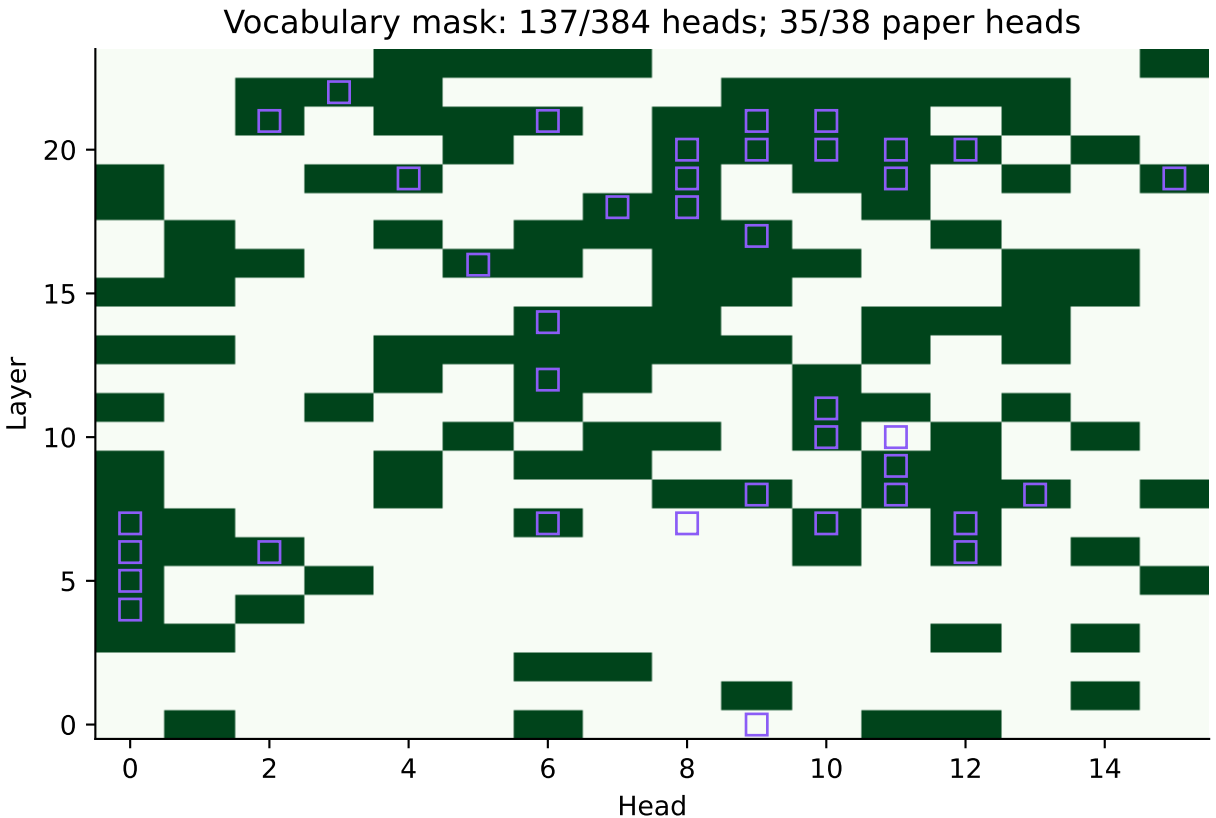
Conditional fidelity and matched random controls

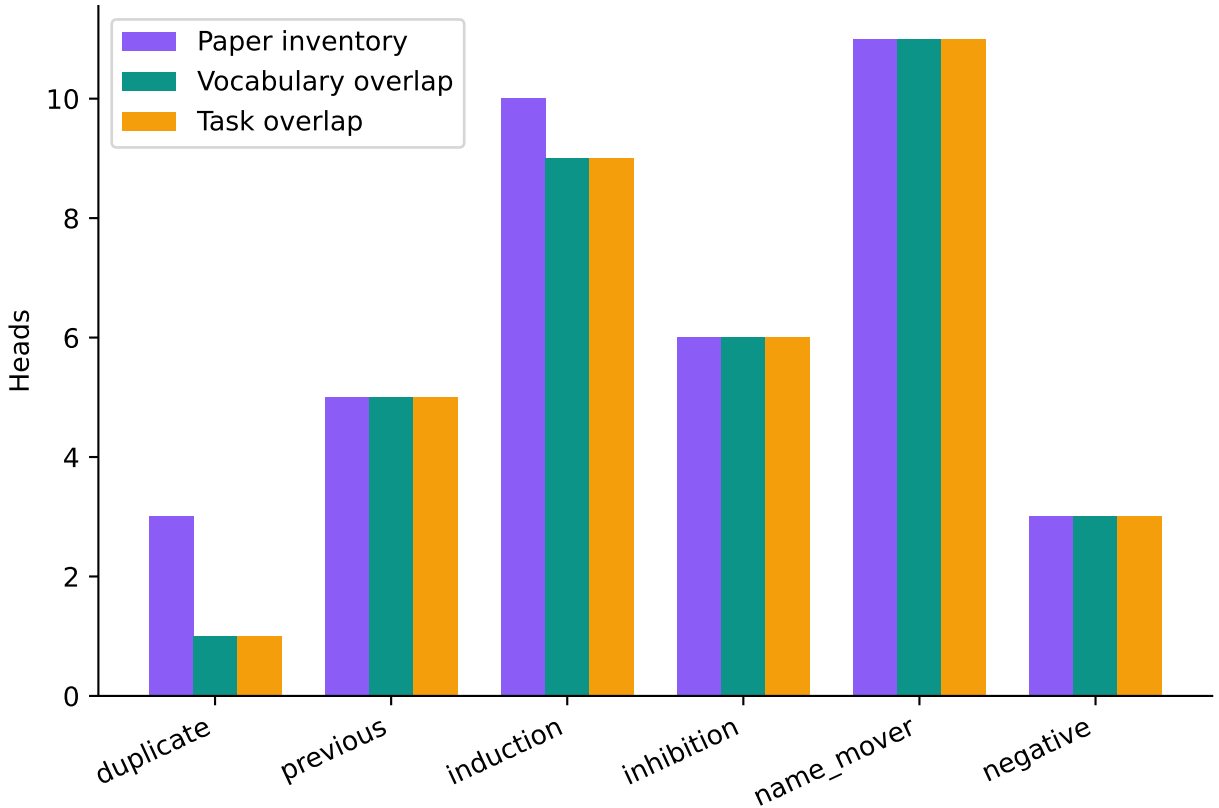




Anchor knockout groups were used during development; random subsets were not. Both variants receive identical knockouts sampled from their shared heads. Fresh test prompts throughout. Random masks match retained head counts at each layer. Finite tests do not prove completeness.

Behavioral fidelity has a circuit-size cost





Green cells: retained whole heads. Purple outlines: published heads, used only for evaluation. Head identity overlap does not establish the same functional role or graph. Anchors derive from the prior independent functional probes.